Announcements

- Everyone getting their marked-up things back?
- Assignment turn-in and Grading (next slide)
- Schedule:

  Today (Lecture): Newton's Laws, Energy, Energy Transfer, Equilibrium
  Tomorrow (Lab): Motion, Energy, and Equilibrium
  Thursday (Discussion)]: Analyzing Enegy Transfer
  - Next Week: Electromagnetic Radiation (Light) and Blackbody Radiation

Assignments: Turn-In and Grading

- Each assignment turned in as single PDF document
- Prepare as if you were going to turn in a hard copy, then scan with *Adobe Scan* app. Ask me for help!

Labs: Turn in:

- all pages with answers, measurements, calculations, etc;
- all data sets produced by hand (large downloaded ones not necessary)
- all plots/graphs produced

Grading for assignments

C $\approx$ 75: You turned it in, mostly complete

B $\approx$ 85: You demonstrated effort, majority correct

A $\approx$ 95: You demonstrated careful effort, thought, and logic; clearly discussed issues with partners and me; mostly correct

Today's Reading Quiz (3 min)

Grading for reading quizzes:

- **80**: You turned it in
- **90**: You demonstrated that you read, have basic ideas
- **100**: Correct answer, demonstrated you internalized ideas (or took careful notes)

On your own sheet of paper...

What does it mean for an object/system to be in equilibrium? How does it relate to "energy in" and "energy out"? What is true about the internal state of the system when in equilibrium?

Turn in to Blackboard as scanned pdf (or image is fine).

Today: Energy and Equlibrium

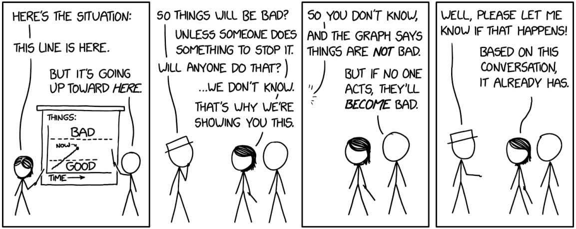
(AKA: Most of first-semester introductory physics in 40 minutes!)

Why are we doing this?

- Last week: temperature of the Earth is increasing.
- Also: (circumstantial) evidence that CO_2 is responsible
- So . . . why do we need more?

A careful *physical* explanation/description will:

- allow us to understand the mechanism for the change,
- give more solid evidence for *attribution* of climate change,
- provide <u>models</u> to predict the future



Motion

We describe the motion an object using three quantities:

- **Position**, x: Where is the object?
- **Velocity**, v: How fast is it "going"? ($\Delta x/\Delta t$)
- **Acceleration**, a: How is the velocity changing? ($\Delta v/\Delta t$)

Each of these has a *direction* associated, so sometimes they are expressed as *vectors*, e.g., $\vec{v}$, $\vec{a}$.

<u>2D Motion (track from above)</u>   <u>1D Motion (on "x" axis)</u>

What is the (average) speed of a car going from GR to Detroit (140 miles) in 2h30m?

First type of Energy: Kinetic Energy

We say that a moving object has *kinetic energy*:

$$\text{Kinetic Energy} = K = \frac{1}{2}mv^2$$

where m is the mass and v is the speed.

<u>Cars (1000 kg) and Trucks (50,000 kg) on the highway</u>

How much faster would a car have to drive to have same K?

Old (wrong) Law of Motion

Aristotle (350 BCE): **What makes it go?**

- Objects are naturally at rest
- A "violent external agent" is needed to make them move
- Once the violent agent stops, the motion ceases

Try pushing a box:

"Makes sense" . . . but the question (and answer) are wrong!

The Law of Motion: "Newton's Second Law"

Newton (1687): **What makes it *change* its motion?**

$$\text{Sum of Forces} = \text{mass} \times \text{acceleration} \quad \rightarrow \quad \vec{a} = \frac{\sum \vec{F}}{m}$$

So: *accelerations* (changes in velocity) are caused by forces (Aristotle's "violent" actions)

Same "pushing the box" experiment, but on ice:

Graphs of Motion (position and velocity vs time)

Three cases are demonstrated:

(i) car at rest; (ii) car pushed and stopped; (iii) car on hill

When does *acceleration* happen? *Why* does it happen?

Other motion concepts (demonstrated)

- Effect of increasing object mass with same applied force.
- Newton's Third Law: Equal but opposite forces in any interaction.

Back to Energy... "Work"

The concept of energy starts with:

Work done by a force gives energy to an object

which is defined as:

$$\text{Work} = \text{Force} \times \text{Distance Applied}$$

Pushing the box, on ice or not:

Work on the box increases its kinetic energy: $W = \Delta K$.

Potential Energy

So far we have:

- Kinetic energy (of motion)
- Work: Transfer of energy by a force

Now a second type of energy: *Potential Energy*

Springs

Gravity

Systems like this can "hold" energy. They have the *potential to do work* with their forces.

Conservation of Energy (I — Mechanical)

From Newton's Laws, one can derive the principle of energy conservation:

> *In a closed mechanical system, energy can be transferred from kinetic to potential, but never lost/destroyed.*

Example with block, spring, hill:

Another type of energy transfer: Heat

Aristotle's box (not on ice) stopped because:

- There is a *frictional* force opposing the motion
- Work done by that frictional force takes energy out of motion, *heating* the block and the surface.

So we now have a second mechanism (in addition to work) of energy transfer: Heat

Internal Energy (Solids)

Where does energy lost to friction go? What does it mean that the block and surface get "heated"?

We can think of solid objects as huge collections of masses connected by springs (these are actually atoms, with chemical/quantum bonds). Frictional-type forces do work on the little particles on the surface — causing spring compression (potential energy) and vibration (kinetic energy) — which then transfers into motion of internal particles.

Internal Energy (Fluids)

For a gas or liquid, the internal energy is simply the sum of the kinetic energy of the particles.

Temperature is a measure of the average kinetic energy of such a system.

Energy Types (Summary)

- Kinetic
- Potential
- Internal (kinetic and potential)
- Chemical (bonds)
- Electrical
- Nuclear

Energy Transfer Types (Summary)

- Work (by force)
- Heating (or conduction)
- Convection
- Radiation

Energy Transfer Diagrams/Analysis